ASSESSMENT TOOL-2

CO2-AT2

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1Q) Define Data mining :-

Data mining is the process of extracting useful, hidden, and meaningfull patterns, relationshipps and knowledge from large volumes of data using statistical, Machine learning, and database Technique.

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Any Two Basic Data mining Tasks :-

⇒ ① classification.

⇒ ② clustering.

Steps in KDD (Knowledge Discovery in databases :-

1. Step ① :- Data cleaning.

Step ② :- Data integration

Step ③ :- Data Selection.

Step ④ :- Data transformation

Step ⑤ :- Data mining

Step ⑥ :- pattern Evaluation

Step ⑦ :- knowledge presentation.

(b) Data cleaning :-

Data cleaning is the process of detecting and Correcting inaccurate, incomplete, duplicate, or inconsistent data.

Two Common Data Quality issues :-

→ ① Missing values
→ ② Noisy data.

Methods To handle missing values

* Ignore tuples
* fill with mean, median, Mode.
* predict using ML.

Methods To handle Noisy data :-

* Binning
* Regression
* clustering.

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② Define data Integration :-

⇒ Data integration is the process of combining data from different Sources into a unified database.

Two challenges :-

① Data Redundancy
② Data inconsistency.

Handling Redundancy :-

* Remove duplicate records.
* use correlation analysis.
* Merge Similar attributes.

Eg :- Student_ID appears in two databases.

Handling inconsistency :-

* Standardize formats
* use validation rules.
* Resolve conflicting.

Eg :- Date :- 03/08/26
2026-08-01
Convert into one Standard.

(b) Covariance analysis :-

Covariance measures how two variables change together

* positive → variables increase together.

* Negative → one increases while other decrease.

*. Zero covariance → No relationship.

Population covariance :-

$$\boxed{Cov(X,Y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y}_i)}{N}}$$

Sample covariance :-

$$\boxed{Cov(x,y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n-1}}$$

(3) Given

A = 2, 3, 5, 4

B = 5, 8, 10, 11

Mean :-

$\bar{x} = 3.5$

$\bar{y} = 8.5$

x	y	$(x-\bar{x})$	$(y-\bar{y})$	product
2	5	-1.5	-3.5	5.25
3	8	-0.5	-0.5	0.25
5	10	1.5	1.5	2.25
4	11	0.5	2.5	1.25

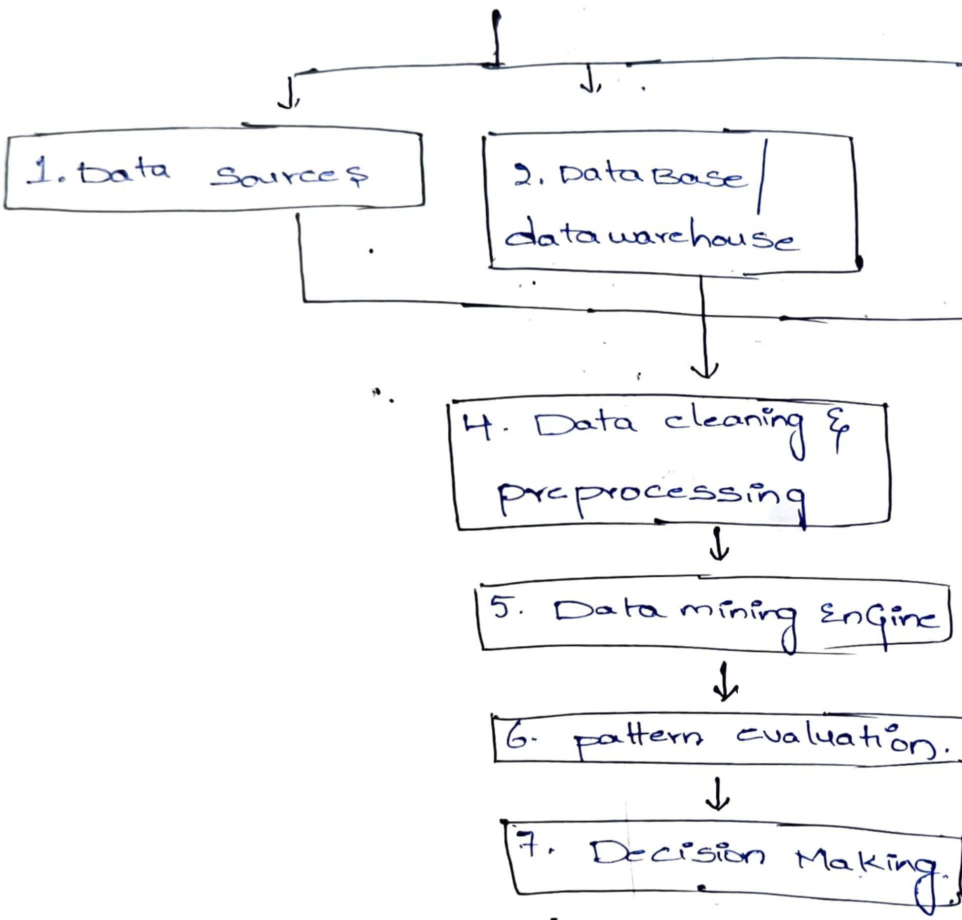
Total - 9

population covariance = $\frac{9}{4}$ = 2.5

Sample covariance = $9/3$ = 3

Conclusion :- Covariance is positive, so both stocks generally rize and fall together.

(b) Architecture of data mining :-



(4) Given

$x = \{4, 5, 6, 7, 9\}$

$y = \{8, 3, 7, 5, 6\}$

Mean $\bar{x} = 6.2$

$\bar{y} = 5.8$.

x	y	$x-\bar{x}$	$y-\bar{y}$	product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

Sum = -1.8 ; population Covariance = $-1.8/5 = -0.36$.

Sample Covariance = $-1.8/4 = -0.45$ //

(b) **Major steps in data preprocessing :-**

1. Data cleaning.
2. Data integration.
3. Data transformation.
4. Data Reduction.
5. Data Discretization.

Pipe line :-

Raw data → Missing value → outlier detection → duplicate removal → feature → clean dataset.

5 (a) **Equal depth Binning :-**

Data set :-

(13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70)

Total = 26 values. ⇒ using 5 bins ≈ 5 values each.

Bin ①

13, 15, 16, 16, 19

Mean = 15.8

Bin ②

20, 20, 21, 22, 22

Mean = 21

Bin ③ :-

25, 25, 25, 25, 30

Mean = 26.

Bin ④ :-

33, 33, 35, 35, 35

Mean = 34.2.

Bin ⑤ :-

36, 40, 45, 46, 52, 70

Mean = 48.17.

Smoothed values :-

48.17, 48.17, 48.17, 48.17, 48.17, 48.17.

(b) Data Discretization :-

* Data discretization is the process of Converting Continuous numerical data into discrete Intervals.

Equal width Binning :-

* divides data into bins of equal numerical range
* Simple to implement
* May produce uneven numbers of records.

Equal frequency Binning :-

* devides data each bin Contains Same no. of Records.
* Better of uneven data.
* Bin range May differ.

Impact on data :-

* Reduce noise and Simplifies data.
* Enhances visualization.